

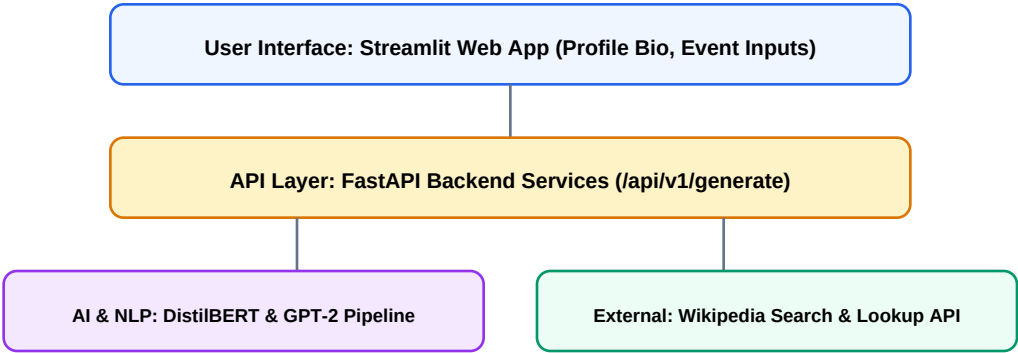
Solution Architecture

Date	29 June 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Solution Architecture

The diagram below maps out the system's Presentation Layer, API Layer, AI Inference & External Integration Layer, and Data Storage Layer.

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides event attendees with text inputs, generate buttons, list views, and voting interface.	Streamlit framework
API Gateway	Handles JSON payload routing, Pydantic type validation, and connects frontend with AI pipelines.	FastAPI, Uvicorn
AI & NLP Services	Performs DistilBERT zero-shot theme extraction and GPT-2 conversational prompt generation.	transformers, torch
Data Persistence	Logs conversation history and user vote feedback in local system folders.	JSON flat files

Three-Tier Layers and Communication Protocols:

Communication between layers is conducted entirely via standardized JSON payloads over HTTP REST protocols. When the user interacts with the Streamlit client, the frontend performs an asynchronous POST request to the local FastAPI server. The backend intercepts the payload, runs validation via Pydantic model contracts, and routes it to the correct service module. The theme extraction module feeds text into the zero-shot classifier, which maps keywords into top candidate themes. These themes, combined with the user's specific interests, are passed to the generator to produce icebreakers. Feedback is saved synchronously on disk to prevent asynchronous thread contentions on files.